Supplemental Data: Rule-Unit Anatomy, Retrieval Loop, Provenance and Detector Baseline

Fangjie Wu, Xingpeng Jiang, Wenlong Zhang, Boran Wang and Jie Gong

This supplement accompanies "Applicability-Aware Rule Retrieval for Regulation-Grounded Construction-Site Visual Screening". It reports the field anatomy of a single provision unit (Fig. S1), the pseudocode of the tool-calling retriever (Algorithm S1), the per-provision provenance of the 42-provision library (Table S1), the dedicated-detector baseline with example cases (Fig. S2), and the unabridged evidence chains of the three worked cases shown abridged in Fig. 2 of the manuscript. The same information is machine-readable in the public artifact repository (Zenodo 10.5281/zenodo.21535053). Nothing here is needed to follow the argument of the manuscript; this file records the detail a reader would need in order to audit or reproduce it.

RULE-UNIT ANATOMY

Fig. S1 shows the typed fields of one provision unit and the auditable chain they expand into. The example is `R-BHV-002`, the hard-hat provision.

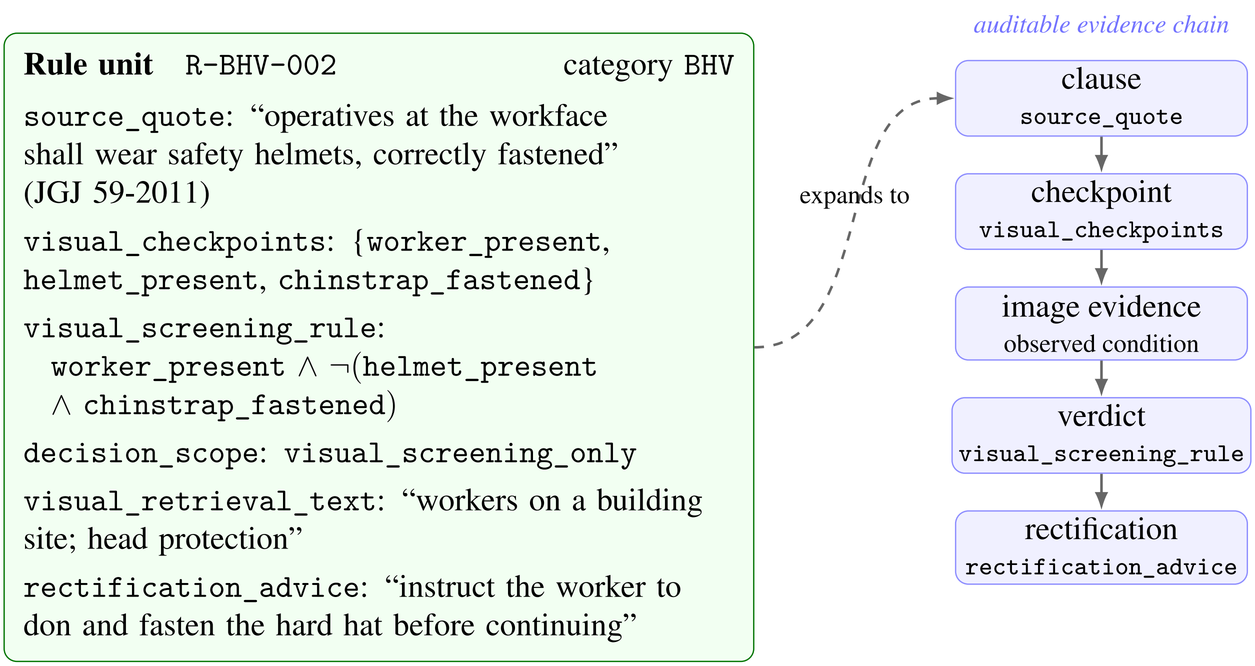

FIG. S1. Anatomy of a provision unit (example `R-BHV-002`, the hard-hat provision). The typed fields (left) carry provenance, atomic perception targets, a Boolean decision formula with an explicit decision scope, retrieval text and remediation. Because the fields are separable, one hit expands deterministically into the five-link auditable chain (right) that a reviewer inspects.

RETRIEVAL LOOP

Algorithm S1 specifies the Stage-2 tool-calling retriever of the manuscript's "Rule retrieval: fixed and tool-calling" section. R3 and R4 differ only in the observation O: R4 additionally receives the raw image on every turn.

Algorithm S1: Bounded tool-calling rule retrieval (R3/R4) with anytime termination. $\textsc{ForceSubmit}(H)$ takes one extra turn with the tool pinned to `submit` and, if that still yields no known `rule_id`, falls back to the ids in H ranked most-hit-first. An iteration-cap hit, a stall, a truncated turn or an unusable submission thus degrades to a best-effort candidate list rather than an empty one.

Input : observation O (cached facts; plus the raw image for R4); provision corpus $\mathscr{C}$; iteration budget N; stall limit L.

Output: a ranked list of candidate `rule_ids`.

```
H ← {}                                   // surfaced rule_id ↦ cumulative grep hits
P ← ∅;  s ← 0;  e ← 0                    // patterns run; stall counter; empty turns
for i ← 1 to N do
    force ← (i = N)                      // final turn pins the tool to submit
    r ← Model(history, tools, force)
    if r emits no tool call then
        e ← e + 1
        if r truncated or e ≥ 2 or force then
            return ForceSubmit(H)
        append retry nudge               // try once more
    else
        foreach tool call c ∈ r do
            if c = submit(ids) then
                K ← {x ∈ ids : x ∈ 𝒞}    // keep known ids
                if K ≠ ∅ then
                    return Rank(K)
                else
                    return ForceSubmit(H)   // unusable submit
            else if c = grep(p) then
                hits ← 𝒞.grep(p)         // matching rule_ids + snippet
                new ← number of ids in hits not yet in H;  update H
                if p ∈ P or new = 0 then
                    s ← s + 1
                else
                    s ← 0
                P ← P ∪ {p}
            else
                return the full provision detail of x   // c = read(x)
        if s ≥ L then
            return ForceSubmit(H)        // marginal-gain early stop
return ForceSubmit(H)                    // safety net; unreachable when N ≥ 1
```

Loop budget. The iteration cap N is over-provisioned rather than binding. Mean iteration counts saturate near 3.3 once $N \geq 4$, because the marginal-gain stop ($s \geq L$) fires before the cap does, so raising N past 4 buys no additional candidates and lowering it to 4 costs nothing measurable. Scalar judge confidence is likewise saturated above 0.8 almost everywhere and is therefore of little use for selective prediction, which is why abstention in this pipeline is structural (`decision_scope` and `need_review`) rather than a threshold on a score. Per-iteration counts are in `results/retrieval/iter_sweep_summary.md` of the artifact repository.

PER-PROVISION PROVENANCE

Table S1 gives the provenance of all 42 provisions; its caption defines the six columns. The standard designations are **GB 55034** (MOHURD 2022), **GB 12523** (MEE 2025), **JGJ 59** (MOHURD 2011), **JGJ 80** (MOHURD 2016), **Hazard 2024** (MOHURD 2024), **SpecOps 2008** (MOHURD 2008) and **SZ 2018** (SZHCB 2018). Some civilized-construction thresholds are Shenzhen local reference values rather than national mandatory limits, and all `source_quote` text in the English runtime library is an author translation, not a verbatim quotation.

TABLE S1. Per-provision provenance of the 42-provision library. *Cat.*: OPN openings, EDG edges, BHV behaviour and protective equipment, CIV civilized-site. *Standard(s)* names the governing document, cited above; "—" marks a provision with no single governing clause. *Prov.*: `std.`, the provision transcribes a clause of that standard; `eng.`, an engineering operationalization, whether or not a standard is also cited; `acc.` and `mgmt.`, additionally derived from accident-investigation findings or from site management requirements. *Pred.*: `clause` when `visual_screening_rule` is identical to the transcribed `normative_rule` and needs no external field, `proxy` otherwise. *Drop*: normative fields the clause requires that a single image cannot supply. *Scope*: the library's `decision_scope`. These last three are read directly off the frozen library, so they cannot drift from the provisions the pipeline runs.

Rule	Cat.	Standard(s)	Prov.	Pred.	Drop	Scope
BHV-001	BHV	GB 55034; JGJ 80	eng.	proxy	2	screen
BHV-002	BHV	GB 55034	eng.; acc.	clause	0	normative
BHV-004	BHV	JGJ 80	std.	clause	0	normative
BHV-005	BHV	JGJ 80	std.	clause	0	normative
BHV-006	BHV	SpecOps 2008	std.	proxy	2	external
BHV-007	BHV	—	acc.	proxy	1	screen
BHV-008	BHV	JGJ 80	acc.	proxy	1	screen
BHV-009	BHV	Hazard 2024	acc.	proxy	2	screen
BHV-010	BHV	SZ 2018	std.	proxy	1	screen
BHV-011	BHV	Hazard 2024	acc.	proxy	1	external
BHV-012	BHV	—	acc.; mgmt.	clause	0	normative
BHV-013	BHV	JGJ 80	eng.	clause	0	normative
CIV-001	CIV	SZ 2018; JGJ 59	std.	proxy	4	screen
CIV-002	CIV	SZ 2018; JGJ 59	std.	clause	0	normative
CIV-003	CIV	SZ 2018; JGJ 59	std.	proxy	2	screen
CIV-004	CIV	SZ 2018	std.	clause	0	screen
CIV-005	CIV	SZ 2018	std.	proxy	5	screen
CIV-006	CIV	SZ 2018	std.	proxy	6	external
CIV-007	CIV	SZ 2018	std.	proxy	4	screen
CIV-008	CIV	SZ 2018	std.	proxy	6	external
CIV-009	CIV	GB 12523	std.	proxy	7	external
CIV-010	CIV	JGJ 59	std.	proxy	8	screen
CIV-011	CIV	SZ 2018; JGJ 59	std.	proxy	1	screen
CIV-012	CIV	SZ 2018; JGJ 59	std.	proxy	1	screen
CIV-013	CIV	SZ 2018	std.	proxy	3	screen
CIV-014	CIV	SZ 2018; JGJ 59	std.	clause	0	normative
CIV-015	CIV	SZ 2018; JGJ 59	std.	proxy	1	screen
CIV-016	CIV	—	eng.	proxy	0	normative
CIV-018	CIV	SZ 2018; JGJ 59	std.	clause	0	normative
EDG-002	EDG	JGJ 80	std.	clause	0	normative
EDG-003	EDG	JGJ 80	std.	clause	0	normative
EDG-004	EDG	JGJ 80	std.	proxy	2	screen
EDG-005	EDG	Hazard 2024; SZ 2018	std.	clause	0	normative
EDG-006	EDG	JGJ 80	std.	proxy	1	screen
EDG-007	EDG	JGJ 80	std.	proxy	8	screen
EDG-011	EDG	JGJ 80	eng.	proxy	1	screen
EDG-012	EDG	SZ 2018	eng.	clause	0	normative
OPN-001	OPN	SZ 2018; JGJ 80	std.	proxy	5	screen
OPN-004	OPN	JGJ 80	std.	proxy	4	screen
OPN-005	OPN	SZ 2018; JGJ 80	std.	proxy	8	screen
OPN-009	OPN	SZ 2018; JGJ 80	std.	clause	0	normative
OPN-011	OPN	JGJ 80	eng.	clause	0	normative

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DEDICATED-DETECTOR BASELINE AND EXAMPLE CASES

An off-the-shelf hard-hat detector (classes {Hardhat, NO-Hardhat}) can address only the hard-hat provision; every other provision in the library is an action or context predicate a closed-set class scheme cannot produce. Mapping a NO-Hardhat box above threshold to a hard-hat-violation claim, the specialist is high-precision and low-recall (P 0.659, R 0.287, F1 0.400), against P 0.925, R 0.614, F1 0.738 for the proposed pipeline on the same provision, an F1 higher by a factor of 1.85. The cause is a closed-set class-scheme mismatch: of 101 hard-hat-violation images, 70 contain only Hardhat boxes, because a binary scheme cannot encode the improper-wearing and attire branches. The detector covers $101/691 = 14.6\%$ of the violation workload against 100% for the pipeline, though its high precision makes it a natural complementary signal. The comparison uses an off-the-shelf model, so it quantifies the class-scheme ceiling of the detector paradigm as deployed without task-specific training, not the ceiling of a detector trained in domain.

Fig. S2 shows three violations, all rated fully relevant in the rectification study, that such a class scheme structurally cannot flag: a relational hazard (a gap between scaffold and structure), an attribute of a worn item (a helmet with the chin strap unfastened, which a binary hard-hat class labels compliant), and a behavior-in-context rule (working at height without a harness).

(a) Scaffold–structure gap (b) Chin strap unfastened (c) No harness at height

FIG. S2. Violations beyond a closed-set detector (faces anonymized).

UNABRIDGED WORKED-CASE CHAINS

Fig. 2 of the manuscript shows three photographs through the screening chain with the clause and evidence text abridged for space. This section reproduces those three chains in full: every candidate the retriever submitted, the complete clause text and screening formula, every checkpoint evidence line with its status, evidence type and confidence exactly as recorded, the verdict and its decidability, the gold status, and the rectification advice. All values are read from the cached headline R4 $\times$ J2 run; nothing was re-run or re-prompted, and the text is reproduced verbatim rather than paraphrased. Generated by `experiments/pipeline/emit_case_chains_tex.py`.

Case (a): correct detection — `img_0115`

Stage 2, candidates retrieved: `R-EDG-002-stair-edge`, `R-EDG-007-guardrail-construction`

Focus provision: `R-EDG-002-stair-edge` — Stairwell, stair landing or flight edge without an installed guardrail

Clause (`source_quote`): "Construction stairwells, stair landings and flight edges shall have guardrails; external stairs shall also be enclosed with vertical netting."

`source_level` / `decision_scope`: Industry standard / `normative_decision_available`

Screening formula: `stair_edge_present == yes AND (guardrail_present == no OR guardrail_stable == no OR (external_stair_present == yes AND outer_stair_net_present == no))`

Stage 5 verdict: `non_compliant`, decidability `decidable`, confidence 1.0

Gold status for this pair: `non_compliant`

Rectification advice: "Install secure guardrails immediately along the open left edge of the stair flight and stair landings to prevent fall hazards."

Stages 3–4, every checkpoint evidence line as recorded:

`stair_edge_present` **satisfied** (direct_visible, conf. 1.0): A concrete stair flight is clearly visible, ascending from bottom-left to top-right.

`external_stair_present` **satisfied** (indirect_visible, conf. 0.8): Background elements such as green safety netting and orange scaffolding suggest an active construction site environment, likely involving external access. *Note:* Context implies external exposure, though strictly internal is possible.

`guardrail_present` **violated** (direct_visible, conf. 1.0): The entire open left side of the stair flight is exposed. There is no continuous horizontal rail or post structure installed along the edge to prevent falls. *Note:* Only temporary loose materials (pipes) are seen near the edge, which do not constitute a compliant guardrail.

`guardrail_stable` **not_visible** (not_visible, conf. 0.0): Since no guardrail is present, stability cannot be assessed.

`outer_stair_net_present` **violated** (direct_visible, conf. 1.0): There is no vertical mesh or netting attached to the open edge of the stairs.

Gold violations on this image: `R-EDG-002-stair-edge`. *Gold-violated but never retrieved:* none.

Case (b): sibling-confusion failure — `img_0188`
Stage 2, candidates retrieved: `R-OPN-001-horizontal-opening-protection`, `R-OPN-009-service-shaft-opening`

Focus provision: `R-OPN-001-horizontal-opening-protection` — Horizontal opening not effectively covered or lacking guardrail/safety-net protection

Clause (`source_quote`): "A horizontal reserved floor opening shall be protected by size: 25-500 mm short edge covered with a load-rated cover and secured; 500-1500 mm covered or fitted with a guardrail, firmly secured; short edge >=1500 mm fitted with a >=1.2 m guardrail sealed by a horizontal safety net with a >=200 mm toe board; the Shenzhen standard simplifies this to a rigid fixed cover for openings <1500 mm."

`source_level` / `decision_scope`: Industry standard / Shenzhen local standard / `visual_screening_only`

Screening formula: `horizontal_opening_present == yes AND (((cover_present == no OR cover_fixed == no) AND (guardrail_present == no OR safety_net_present == no)) OR protection_intact_and_stable == no)`

Stage 5 verdict: `non_compliant`, decidability `decidable`, confidence 1.0

Gold status for this pair: `not annotated`

Rectification advice: "Immediately place the cover panel back onto the opening to seal it. Ensure the cover is rigid, load-rated, and securely fastened (fixed) so it cannot be easily displaced. If the cover cannot be secured, install a temporary guardrail system."

Stages 3–4, every checkpoint evidence line as recorded:

`horizontal_opening_present` **satisfied** (direct_visible, conf. 1.0): A rectangular void/opening is clearly visible in the concrete floor/landing next to the stairs.

`cover_present` **satisfied** (direct_visible, conf. 1.0): A flat, weathered panel (likely plywood or wood) is visible resting on the floor surface immediately next to the opening.

`cover_fixed` **violated** (direct_visible, conf. 1.0): The cover panel is lying flat on the ground beside the hole rather than covering it. It is not secured over the opening. *Note:* The cover is displaced and does not protect the opening.

`protection_intact_and_stable` **violated** (direct_visible, conf. 1.0): The opening is completely exposed without any effective barrier preventing access to the void. *Note:* The protective measure has failed due to the cover being moved aside.

`guardrail_present` **violated** (direct_visible, conf. 1.0): No guardrails are installed around the perimeter of the opening.

`safety_net_present` **violated** (direct_visible, conf. 1.0): No safety nets are visible.

`toe_board_present` **violated** (direct_visible, conf. 1.0): No toe boards are visible.

Gold violations on this image: `R-EDG-002-stair-edge`, `R-OPN-004-vertical-and-low-sill-opening`. *Gold-violated but never retrieved:* `R-EDG-002-stair-edge`, `R-OPN-004-vertical-and-low-sill-opening`.

Case (c): abstention — `img_0263`
Stage 2, candidates retrieved: `R-OPN-001-horizontal-opening-protection`, `R-CIV-003-material-stacking-labeling`, `R-CIV-005-spoil-waste-overheight-uncleared`

Focus provision: `R-CIV-005-spoil-waste-overheight-uncleared` — Spoil or waste stacked over height or left uncleared beyond 48 hours

Clause (`source_quote`): "Spoil stacking height <=1.5 m and removed within 48 h; waste brick/concrete storage pits limited to 1.5 m, removed within 48 h after completion."

`source_level` / `decision_scope`: Shenzhen local standard / `visual_screening_only`

Screening formula: `waste_or_spoil_present == yes AND spoil_grossly_overheight == yes`

Stage 5 verdict: `need_review`, decidability `partially_decidable`, confidence 0.6

Gold status for this pair: `not annotated`

Rectification advice: "Immediately measure the height of the waste pile. If it exceeds 1.5 meters, reduce the height immediately. Verify the date the waste was deposited; if it has been more than 48 hours, arrange for immediate removal. Ensure all waste is stored in designated areas and kept orderly."

Stages 3–4, every checkpoint evidence line as recorded:

`waste_or_spoil_present` **satisfied** (direct_visible, conf. 1.0): A large, chaotic pile of mixed construction debris is clearly visible in the immediate foreground, consisting of wooden boards, plywood sheets, metal scaffolding tubes, and fasteners. *Note:* Subject is clearly present and occupies a significant portion of the lower frame.

`spoil_grossly_overheight` **need_review** (indirect_visible, conf. 0.7): The pile extends vertically well above the bottom edge of the frame. Comparing the height of the debris stack to the vertical structural columns and the window openings suggests the pile is substantial, likely exceeding 1 meter, though exact measurement is not possible. *Note:* Visual estimation indicates potential over-height, but definitive confirmation of >1.5m requires measurement.

Missing information recorded: Exact pile height in meters (requires manual measurement); Time elapsed since waste was deposited (requires site logs).
Gold violations on this image: `R-CIV-003-material-stacking-labeling`, `R-EDG-002-stair-edge`. *Gold-violated but never retrieved:* `R-EDG-002-stair-edge`.